

NOISFERATU

Infinite textures and digital mayhem

NOISFERATU is a compact handheld generative texture synthesizer.

45 algorithms across 5 banks produce crackling textures, noisy scapes, evolving drones, harmonic blips and digital chaos. Shaped but never fully controlled by four knobs.

They generate endlessly evolving textures, tones and rhythmic patterns. Designed to surprise, resist repetition and never quite settle. Dial in a crackling drone, a stuttering melody, or pure digital chaos.

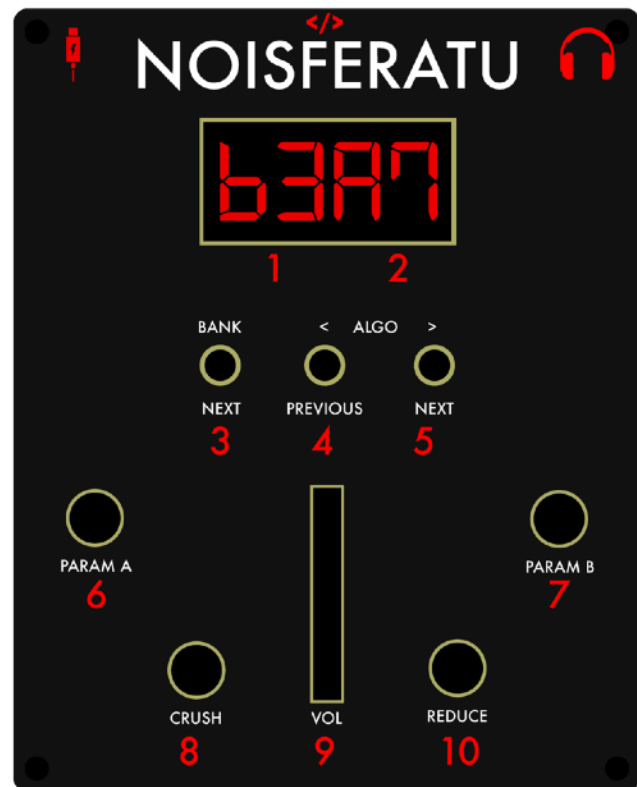
It uses and abuses creative approaches like generative wavetables, BitBend address manipulation, probability gates, bitwise logic operations and wild combinations of these.

Powered via USB C, you can hook it up to a computer, a smartphone charger or a power bank.

This instrument focuses on creative, generative textures, unusual patterns, and playable patches.

From harmonic drones to bit crushed mayhem — always evolving, always surprising.

- 🔌 USB power 🎧 Headphone out
- </> Flash firmware
- 1 Display: Bank 1 to 5
- 2 Display: Algo 1 to 9
- 3 Next bank (Long press Display ON/ OFF)
- 4 Previous algo
- 5 Next algo
- 6 and 7 Param A and B control
two unique parameters per
algorithm. Check the bank details
or just turn, listen and explore.
- 8 Global bitcrush
- 9 Main volume
- 10 Global sample rate reduction



Banks - Short overview

Bank 1 — Generative Wavetables

Glitchy bursts and evolving soundscapes. From sparse crackles to dense digital chaos.

Bank 2 — Noisy Textures

Noise, dust and rhythmic grain. From windy hiss to clicks, crackles and bursts.

Bank 3 — BitBend

Warped, stuttering and unpredictable. Always shifting, never settling.

Bank 4 — Blips & Tones

Melodies, chords and rhythmic blips. The most tonal bank.

Bank 5 — Logic Disorder

Two oscillators fighting each other. Pure digital mayhem.

Display - ON/ OFF (time out)

The Noisferatu uses a 4 digit 7 segment display.

By pressing the BANK button for 2 seconds you can toggle between display ON and display OFF.

Choosing ON the display will always display the current bank and algo.

If you choose OFF the display turns off after 500 milliseconds after you used one of the buttons.

Banks - Approaches

Bank 1 — Generative Wavetables

Instead of using fixed wavetables, the core idea in these algorithms is to define rules that create a wavetable at each start of the algo. These rules define what gets written in the wavetable. For example, noise / latched noise, how much silence chunks are inserted, if tonal elements shall be added, how these elements are spread across the wavetable.

The great thing is that at each start of such an generative algo the wavetable is never exactly the same as in the previous generation, but they still sound familiar, since they follow the same core rules of wavetable creation.

Bank 2 — Noisy Textures

This bank started with the Dust algo, since i love adding little crackles to compositions. From there i tried different approaches to noise, wind, crackles and rhythmic grains. At its core most of this algos use simple white noise and add different operations and processing for different effects.

Bank 3 — BitBend

This bank uses the same generative wavetable approach as in bank 1, but takes it one step further by adding „BitBending“.

The BitBend technique: ****manipulate the read address before accessing the buffer**** is inspired by SoundScaper's "Clock Address Lines (SoundScaper is a great software for ios by Igor Vasiliev).

Instead of changing **what** is in the buffer, you change **where** you read from on each sample. Four modes: SET_0 (force bits to 0 → loops in lower addresses), SET_1 (force bits to 1 → jumps forward), XOR (flip bits → chaos), HOLD (freeze captured bits → locked loops). Algorithms combine 1–4 modes on different bit ranges simultaneously, with a clocked Bernoulli gate randomly evolving the bit positions. This means you can pretty much listen to a sound and it will keep evolving and changing, since the clocked Bernoulli gate probabilistically reassigns which bit positions are affected over time.

Bank 4 — Blips & Tones

After coding lots of crackles, glitches and noises i felt the Noisferatu needed one more tonal bank. With bitcrush and sample rate reduction you can easily turn these into crackling glitches anyway. The approaches here are in some cases simple generative music and algorithmic composition methods like random walks, probabilities in other cases phase shifting amplitude modulations or just a ring modulation.

Bank 5 — Logic Disorder

Pure bitwise operations on live oscillator values. It's doing what a CMOS logic gate would do — but on 10 bits simultaneously, once per sample, 16000 times per second. This can create complex harmonic content and chaotic behavior. It is super fun to play.

Banks in more detail

Bank 1: Generative Wavetables

Buffer-based textures using generative wavetables. All variants regenerate on algorithm switch, each shaped by its own musical rules. GW17 and GW18 use advanced techniques (harmonic drones and quad bit manipulation).

#	Name	Description	Param A	Param B
1	Sparse Glitchy	Clicks and clusters. Rare noise chunks (10%), 1 triangle blip (220–7kHz), chunked silence injection.	speed	silence prob (max 20%)
2	Dense Microglitch	Rhythmic clicks. Tiny chunks (2/5/21 samples), denser noise (3%), high blip (2–2.2kHz).	speed	silence prob (0–100%)
3	Spacey Pulses	A spanner in the works. 33-sample chunks, 25% noise, no blip, chunked silence injection.	speed	silence prob (max 10%)
4	Random Jump Glitch	Grainy textures. 16/4/32 chunks, 50% noise, random buffer jump on silence.	speed	silence prob
5	Wandering Window	R2D2 on acid. Noise/silence chunks + blip (600–2.5kHz), window randomly walks ± 20 samples.	speed	walk rate
6	Manual Window + Spray	Failed hard drive. Moderate density + blip (700–3.5kHz), fixed $0.5\times$ speed, ± 30 sample spray offset.	window pos	window size
7	Noise or Saw Window	Stuttering tones. Chunks are 50/50 noise or saw (30–1kHz) + blip, wandering window playback.	speed	walk rate
8	Chord Loop	Broken chord and chiptunes. 30% noise crackles + harmonic triangle pairs ($\text{root}/2\times/3\times/4\times$), partial regen every 3s, pot2 stutter/freeze.	octave mult	stutter prob
9	BitBend Quad	Angry fax machine. Super-latched sparse noise, $3\times$ big blips (600–900Hz), 4-layer address manipulation (XOR+HOLD+SET_0+SET_1).	speed	bit clock

Bank 2: Noisy Textures

Noise-based processing, filtering, and rhythmic textures.

Slot 9 uses a PROGMEM sample.

#	Name	Description	Param A	Param B
1	Latched Noise	Broken machinery. S&H noise with probability gate.	clock prob	clock rate
2	Dust	Dust and tape hiss like noise. Sparse clicks + 1-pole lowpass filter.	lowpass cutoff	density
3	FMnoise	Broken modem. S&H noise modulates phase increment of sawtooth.	base freq	noise S&H rate
4	Noisegates	Rhythmic crackles. Two triangle oscillators gate noise; osc2 has random walk on freq.	base freq	walk step
5	Saw Clicks	Rhythmic creaking. Two detuned saws, averaged.	saw1 freq	saw2 freq
6	Noise NOR Noise	Crushed rhythmic drone. 2 S&H noise sources.	S&H1 rate	S&H2 rate
7	Dust Burst	White noise bursts. Dust + Bernoulli-gated noise bursts (15% duty cycle).	burst clock rate	dust density
8	Highpass Noise	Wind and shaker noises. Bit-masked highpass noise + triangle LFO amplitude modulation.	AM depth (inverted)	AM rate
9	Crackles	Crackle texture. Sample, random window jumps with chunked silence. Speed 0.04–2× for pitch/texture.	speed/pitch	window size (200– 4000 samples)

Bank 3: BitBend Wavetables

The BitBend technique: ****manipulate the read address before accessing the buffer**** — inspired by SoundScaper's "Clock Address Lines." Instead of changing ***what*** is in the buffer, you change ***where*** you read from on each sample. Four modes: SET_0 (force bits to 0 → loops in lower addresses), SET_1 (force bits to 1 → jumps forward), XOR (flip bits → chaos), HOLD (freeze captured bits → locked loops). Algorithms combine 1–4 modes on different bit ranges simultaneously, with a clocked Bernoulli gate randomly evolving the bit positions.

#	Name	Description	Param A	Param B
1	Chaos	Bouncing fractals. Dense noise+saw chunks. 1× generative, auto-picks mode randomly.	playback speed	bit clock rate
2	Sparse	Ping Pong ball. Sparse Glitchy. 1× mode, pot2 selects mode directly (4 options).	playback speed	mode selection
3	Dual	Circuit bent toy. Spacey Pulses. XOR (low bits) + HOLD (high bits), 50/50 Bernoulli gate.	playback speed	bit clock rate
4	Freeze	Stuttering texture. Wandering Window. SET_0 (low bits) + HOLD (high bits), fixed 23Hz clock.	playback speed	bit clock rate
5	Ping	Stuck plotter. Custom sparse noise + high-freq tris (3–4kHz). XOR (low bits) + SET_1 (high bits), 50/50 Bernoulli gate.	playback speed	bit clock rate
6	Mirror	Pitch fluctuation. Custom noise+saw with freq sweep. SET_0 (low) + SET_1 (high) — conflicting forces lock address to middle range.	playback speed	bit clock rate
7	Triple	Chiptune medley. XOR (bits 0-2) + HOLD (mid) + SET_0 (bits 6-9), 50/50 Bernoulli gate.	playback speed	bit clock rate
8	Sweep	Tones and Stutters. Dual HOLD at two speeds — slow clock on low bits, fast clock on high bits.	playback speed	bit clock rate
9	Triple B	Moody signals. Custom sparse noise + 3× high-freq blips (4–6kHz). SET_0 (bits 0-1) + HOLD (bits 2-4) + SET_1 (bits 5-9), 50/50 Bernoulli gate.	playback speed	bit clock rate

Bank 4: Blips & Tones

Melodic and rhythmic generators.
Enveloped oscillators, scales, ring mod, Bernoulli gates.

#	Name	Description	Param A	Param B
1	Random Triangle	Tonal blips. Enveloped triangle, 50% skip trigger, random freq 250–2.78kHz.	trigger rate	env decay
2	Harmonic Tris	Drones and harmonic clusters. 3 triangles (1×, 3×, 4×) + 3 LFOs with irrational polyrhythm ratios (1, 4:3, $\sqrt{3}$).	LFO1 rate	LFO2 rate
3	Fast Triangle	Modulated distorted groovy tunes. ~960 BPM clock, 75% skip, random bitcrush per trigger (3–12 bits).	freq	decay
4	Phrygian Tri	Walk across harmonic tones. Triangle walks 8-note Phrygian scale via random walk, burst-modulated trigger rate, 50ms decay.	burst speed	root freq
5	Ring Mod	Inharmonic sidebands for nastily beautiful drones. Two triangles multiplied <code>`(tri1×tri2)>>9`</code> .	osc1 freq	osc2 freq
6	Noise OR Square	Broken slot machine. S&H noise bitwise OR'd with square wave.	noise S&H rate	square freq
7	Major Tris	Fairy Tale. Fixed 220Hz major chord (root + maj3rd + perf5th), nested LFO AM on each voice.	LFO1 rate	LFO2 rate
8	Bernoulli Minor 7th	Rich harmonic relations. 5Hz clock → 50/50 gate → two Bernoulli note gates (220/330 Hz vs 264/396 Hz = minor 7th chord).	gate1 prob	gate2 prob
9	Pentatonic Blips	Bouncy harmonics. Random major pentatonic notes from 220Hz, 50/50 Bernoulli clock gate.	clock speed	env decay

Bank 5: Logic Disorder

Pure bitwise operations on oscillators.
These algos create complex harmonic content.

#	Name	Description	Param A	Param B
1	Three Cascaded Squares	Distorted mayhem. Three squares. AM chain: sq1 modulates sq2, sq2 modulates sq3. Each square wave controls the volume of the next.	sq2 freq	sq3 freq
2	NOR Square	Noisy drones. Two squares. Silent when either wave is active, loud only in the gaps.	sq1 freq	sq2 freq
3	Tri OR Saw	High and low drones. Triangle + Saw. Combines both waves, emphasising shared loud moments.	tri freq	saw freq
4	Tri NOR Tri	Chirping madness. Two triangles. Only speaks in the silence between both waves.	tri1 freq	tri2 freq
5	Tri XOR Tri	Noise and harmonics. Two triangles. Active only where the two waves disagree — crackle and chaos.	tri1 freq	tri2 freq
6	Square XNOR	Noisy beating. Two squares. Active only where the two waves agree — dense and buzzy.	sq1 freq	sq2 freq
7	Square NAND	Transmission and static interferences. Two squares. Silent only when both waves are fully active simultaneously.	sq1 freq (0.1–50Hz)	sq2 freq
8	Two Saws	Rich sonic scapes. Two sawtooth waves. The two waves mixed together — detuning creates movement.	saw1 freq	saw2 freq
9	Square OR Square	Marching toys. Two squares. Active whenever either wave is active — full and aggressive.	sq1 freq	sq2 freq